

Long Term Behavior - Simple Case [Lecture 38]

from Lec-37:-

for distinct eigen values: gen sol to $\dot{\vec{x}} = A\vec{x}$ is given by

$$\vec{x} = \sum_{i=1}^n \vec{u}_i e^{\lambda_i t}$$

$$\downarrow$$

$$R_n \rightarrow R^n$$

$$\downarrow$$

 $\vec{x}$ is a Linear Combination of eigen vectors of A .

Coefficients of Linear Comb

are determined by the eig values

more precisely $e^{\lambda_i t}$ if at least one $\lambda_i > 0$:-

$$e^{\lambda_i t} \rightarrow \infty \text{ as } t \rightarrow \infty \text{ if } \lambda_i > 0.$$

$$\downarrow$$

$$\text{Solution } \vec{x}(t) \rightarrow \infty \text{ as } t \rightarrow \infty$$

Vector

Not a precise notation $\rightarrow$ used to imply that at least one elem
approaches $\infty \Rightarrow$ norm of $\vec{x} \rightarrow \infty$.if $\lambda_i < 0 \forall i$:-

$$e^{\lambda_i t} \rightarrow 0 \text{ as } t \rightarrow \infty \forall i$$

$$\downarrow$$

$$\vec{x}(t) \rightarrow \vec{0} \text{ as } t \rightarrow \infty$$

when studying nonlinear system of ODE's $\xrightarrow[\text{an Equilibrium}]{\text{linearize at}}$ $\dot{\vec{x}} = A\vec{x}$ $\xrightarrow{\text{check if}}$ eigen values of $A < 0$ ($\lambda_i < 0$)

if yes, the nonlinear system is stable at that Equilibrium.

Diagonalization

Let eigenvalues of A be real, negative and distinct. Let the eigen values be arranged such that

$$|\lambda_1| > |\lambda_2| > |\lambda_3| \dots > |\lambda_n|$$

$$\downarrow$$

$$e^{\lambda_1 t} < e^{\lambda_2 t} < \dots < e^{\lambda_n t}$$

← Since they rep the Coefficients of Linear Combination

$$\downarrow$$

$$\vec{x}(t) \rightarrow \text{closer to } \vec{0}_n$$

(Need to be careful

with this as there is some dependence of the eventual sol to IVP to the initial condition).

$$\downarrow$$

$$\text{as } t \rightarrow \infty, \vec{x}(t) \rightarrow \vec{0}$$

(trajectory that is close to $\vec{0}_n$).

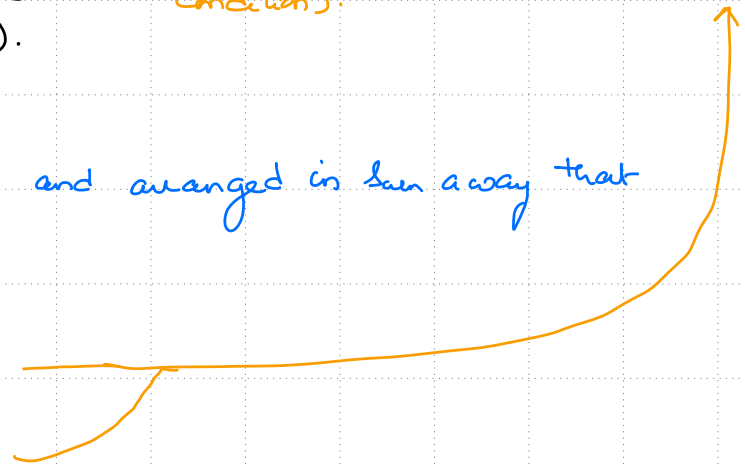
when λ_i 's are distinct and atleast one $\lambda_i > 0$ and arranged in such a way that

$$\lambda_1 > \lambda_2 \dots > \lambda_n$$

Largest Coefficient of LC: $e^{\lambda_1 t}$

$$\vec{x}(t) \rightarrow \text{closer to } \vec{0}_1$$

Trajectory is relatively close to $\vec{0}_1$ as $t \rightarrow \infty$



Diagonalization

Let $\dot{\vec{x}} = A\vec{x}$ $\xrightarrow{\textcircled{1}}$ have real and distinct eigen values.

$\downarrow$
the eigen vectors

$\vec{v}_1, \dots, \vec{v}_n$ are linearly independent

The matrix $V = [\vec{v}_1 \dots \vec{v}_n] \in \mathbb{R}^{n \times n} \rightarrow$ full Rank
 $\downarrow$
 V^{-1} exists.

The product $AV = A[\vec{v}_1 \dots \vec{v}_n] = [\lambda_1 \vec{v}_1, \dots, \lambda_n \vec{v}_n]$

$= VD$ where $D = \text{diag} \{\lambda_1, \dots, \lambda_n\}$

$\therefore AV = VD \Rightarrow A = VD V^{-1} \xrightarrow{\textcircled{2}}$

using $\textcircled{2}$, $\textcircled{1}$ can be expressed as $\dot{\vec{x}} = VD V^{-1} \vec{x}$

$$\underbrace{V^{-1} \dot{\vec{x}}}_{\vec{y}} = \underbrace{D V^{-1} \vec{x}}_{\vec{y}} = V^{-1} \vec{x} \Rightarrow \dot{\vec{y}} = V^{-1} \dot{\vec{x}}$$

So a new
decoupled system
is obtained $\left\{ \begin{array}{l} \dot{\vec{y}} = D \vec{y} \end{array} \right.$